1 Introduction

Since the launch of ChatGPT in autumn 2022, generative AI has developed quickly across text, code, image, and other creative tasks. This has raised a clear labour-market question: why have such large technical advances not yet produced large aggregate employment effects? Earlier computerisation produced visible structural change, including the occupational polarisation documented by Autor & Dorn (2013). One might therefore expect large language models to generate similarly visible disruption, especially because Eloundou et al. (2024) estimate that a large share of workers hold occupations with tasks exposed to LLMs.

Yet recent evidence is more muted at the aggregate level. The task framework of Acemoglu & Restrepo (2019) predicts that automation can both displace tasks and create new demand, so the net employment effect is ambiguous. Consistent with this, Massenkoff & McCrory (2026) find no systematic unemployment increase in highly exposed occupations, while Brynjolfsson et al. (2025) find sharper losses for early-career workers rather than broad occupational collapse. This suggests that the main adjustment may be hidden inside occupations.

I argue that aggregate null results can mask offsetting compositional shifts within occupations. AI may displace junior workers, whose tasks overlap more with current AI capabilities, while augmenting senior workers, whose expertise-intensive tasks remain complementary to AI. Since both groups sit inside the same occupation codes, junior displacement and senior augmentation may partly cancel in occupation-level employment measures. This idea is consistent with the seniority-biased effects documented by Lichtinger & Hosseini Maasoum (2025) and with evidence that early-career workers are especially exposed (Brynjolfsson et al., 2025).

This framing also helps connect two strands of evidence that otherwise look separate. Studies of aggregate occupation-level employment often find small or delayed effects,

while firm- or worker-level studies increasingly point to pressure on entry-level work. The contribution of this essay is to treat these as related facts rather than competing results: the same occupation can experience junior displacement and senior reallocation at the same time.

The theoretical contribution is to extend the expertise framework of Autor & Thompson (2025) by relaxing within-occupation worker homogeneity. In my partial-equilibrium CES model, junior and senior workers have comparative advantage in different tasks. AI substitutes for junior labour in low-expertise tasks and complements senior labour through task-level production. The model predicts a falling junior employment share, rising senior marginal products, ambiguous net occupation-level employment, and wage movements that may be hard to detect if quantity adjustment is stronger than price adjustment.

The empirical contribution is to test this within-occupation mechanism directly, rather than treating an occupation as a single homogeneous labour input. This is where individual-level position histories are especially useful for the research question studied here.

I test these predictions using UK position data from Revelio Labs matched to the LLM exposure scores of Eloundou et al. (2024). The empirical design treats the release of ChatGPT in November 2022 as a shock and uses cross-occupation variation in exposure. I find a sustained decline in the junior share of new positions in highly exposed occupations. I also study career mobility as a related adjustment margin, addressing the call by Brynjolfsson et al. (2025) and Klein Teeselink (2025) for more worker-level evidence. Senior workers in exposed occupations become more mobile, while junior workers who transition experience larger seniority gains; this estimate is stable after Heckman correction. I find no clear wage effects in the Revelio predicted salary data and no evidence of occupation switching away from exposed occupations.

The remainder of this essay proceeds as follows. Section 2 develops the theoretical framework. Section 3 describes the data and identification strategy. Section 4 presents the empirical results on compositional shifts and career mobility. Section 5 discusses limitations and situates the findings within the broader literature. Section 6 concludes.

2 Theoretical Framework

Autor & Thompson (2025) introduce an expertise framework in which the effect of automation depends on which tasks are automated. When automation replaces inexpert tasks, the remaining task bundle becomes more expertise-intensive; wages may rise, but employment can fall because fewer workers meet the threshold. Their framework assumes that workers inside an occupation are homogeneous: each worker must perform all non-automated tasks and, conditional on meeting the expertise threshold, is equally productive.

This assumption is useful for their between-occupation analysis, but it rules out the within-occupation margin studied here. I keep the idea that occupations bundle tasks of different expertise levels, but allow junior and senior workers to specialise inside the same occupation. The motivation comes from knowledge-hierarchy models such as Garicano & Rossi-Hansberg (2006), where less experienced workers handle more routine problems and experienced workers solve harder ones. This gives a production-side reason why the same occupation code can contain different task portfolios. I use Revelio seniority as the empirical proxy for this distinction.

The model is deliberately partial equilibrium. I do not model how the whole economy reallocates workers across occupations, as in Autor & Thompson (2025). Instead, I ask what happens inside a given occupation when AI becomes more capable. This narrower setup is useful because the empirical analysis also works at the occupation-by-time and

worker-transition levels. It allows the model to focus on the junior–senior composition margin without needing to explain all aggregate employment changes.

2.1 Model Setup

Consider a single occupation in partial equilibrium. Output Y is produced by combining three task types via CES aggregation:

$$Y = \left[\beta_L y_L^{\frac{\sigma-1}{\sigma}} + \beta_H y_H^{\frac{\sigma-1}{\sigma}} + \beta_G y_G^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{1}$$

where $0 < \sigma < 1$ is the within-occupation task substitution elasticity and $\beta_L, \beta_H, \beta_G > 0$ are task share parameters, with $\beta_L + \beta_H + \beta_G = 1$. The parameter σ is distinct from the between-occupation elasticity λ in Autor & Thompson (2025).

Following Autor & Thompson (2025), tasks fall into three categories:

- Task L: low-expertise activities, including routine operations that are readily automatable by AI.
- Task H: high-expertise activities, including context-dependent judgement, managerial decision-making, and other functions requiring accumulated experience.
- Task G: generic support tasks assigned to junior workers, including common execution-oriented activities that remain non-automated.

The generic task allows junior workers to remain useful even when AI substitutes for part of their task bundle. This matters for generative AI because exposure is high in many white-collar occupations (Webb, 2020), where junior workers often perform both routine cognitive tasks and support work. Without task G, the model would predict too stark a form of displacement; with it, the junior share can fall without junior employment disappearing entirely.

Each task type is produced as follows:

$$y_L = A \cdot M + a_L \cdot n_J^L, \quad y_H = b_H \cdot n_S, \quad y_G = c \cdot n_J^G \tag{2}$$

In task L, AI capital M (rented at rate r) and junior labour n_J^L are perfect substitutes, with A denoting AI capability. This is the displacement channel. Task H requires

senior workers because it involves judgement and accumulated experience. This is the augmentation channel: AI raises the value of complementary high-expertise tasks. Task G is junior-assigned support work; I abstract from senior production of G because senior time has a higher return in task H.[1]

Junior workers allocate their labour between tasks L and G subject to $n_J^L + n_J^G = n_J$. At the occupation level, labour supply responds to wages with constant elasticity:

$$n_J = \varphi_J w_J^{\varepsilon}, \quad n_S = \varphi_S w_S^{\varepsilon} \tag{3}$$

where $\varepsilon > 0$ captures workers' ability to enter or exit the occupation in response to wage changes.

2.2 Equilibrium and Comparative Statics

The competitive firm's cost minimisation yields three conditions; full derivations are in Appendix A.1 and Appendix A.3.

No-arbitrage in task L. Since AI and junior labour are perfect substitutes, the firm employs both only when their unit costs are equalised:

$$w_J = \frac{a_L \cdot r}{A} \tag{4}$$

As AI capability A rises, the junior wage w_J falls.

Internal allocation of junior labour. Equalizing the marginal revenue product of junior labour across tasks L and G yields $y_G/y_L = \gamma$, where $\gamma \equiv (\beta_G c/\beta_L a_L)^{\sigma}$ is independent of A.

Senior augmentation via complementarity. Senior wages satisfy:

$$w_S = p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H \tag{5}$$

The term $(Y/y_H)^{1/\sigma}$ is the complementarity channel. Holding senior labour fixed, higher AI capability raises y_L and total output Y, which increases the marginal revenue product

[1] This restriction is equivalent to assuming that the senior return in task H exceeds the return seniors would obtain from generic support work. See Appendix A.2.

of task H and therefore w_S. The resulting rise in senior labour dampens but does not reverse this outward shift in senior demand.

Comparative statics. A rise in A lowers w_J and, with elastic labour supply, reduces n_J. Within junior labour, workers move away from task L toward task G, but total junior employment still falls. At the same time, the increase in y_L raises senior marginal products through CES complementarity, increasing w_S and n_S. The combined effect is a lower junior employment share. The net effect on total occupation-level employment remains ambiguous because the junior decline and senior increase work in opposite directions.

2.3 Testable Predictions

The static model formally delivers four implications:

1. **Compositional shift:** In occupations with higher AI exposure, the junior employment share declines following the AI shock, as junior workers are displaced from task L and exit the occupation.
2. **Senior augmentation:** Senior workers in high-exposure occupations experience rising marginal products, wages, and labour demand through CES complementarity.
3. **Wage adjustment:** Junior wages decline and senior wages rise. If the occupation-level supply elasticity ε is large, wage changes may be small relative to employment changes, but the model still predicts price adjustment.
4. **Aggregate ambiguity:** The net occupation-level employment effect is indeterminate in sign, consistent with the aggregate null results documented in the existing literature.

Because junior workers retain a role in task G, the model predicts partial rather than total displacement. The model is static and does not include search, matching, or skill accumulation, so the career-mobility results below should be read as complementary evidence rather than direct model tests.

3 Data and Identification Strategy

3.1 Data and Specification

I use individual-level position data from Revelio Labs, which aggregates UK LinkedIn profiles. The sample contains 23,305,509 workers and 50,742,348 position records active at some point between January 2019 and December 2025. For each position, I observe the O*NET occupation code, start and end dates, predicted salary, company id, seniority level, and within-individual position number. Table 1 shows the seniority scale.

Seniority	Level	Examples
1	Entry	Accounting Intern, Software Engineer Trainee, Paralegal
2	Junior	Account Receivable Bookkeeper, Junior Software QA Engineer, Legal Adviser
3	Associate	Senior Tax Accountant, Lead Electrical Engineer, Attorney
4	Manager	Account Manager, Superintendent Engineer, Lead Lawyer
5	Director	Chief of Accountants, VP Network Engineering, Head of Legal
6	Executive	Managing Director (Treasury), Director of Engineering (Backend Systems), Attorney (Partner)
7	Senior Executive	CFO, COO, CEO

Table 1: Revelio Labs seniority classification

I define a worker as *junior* if the position has seniority level 1 or 2.[2] This corresponds to Entry and Junior roles and captures 58.9 per cent of the sample. These workers typically perform execution-oriented tasks under supervision. From the Associate level onward, workers have more discretion and are closer to the senior task type in the model. Lichtinger & Hosseini Maasoum (2025) validate this measure by showing that job-title keywords align with expected seniority differences and remain stable around the diffusion of generative AI.

[2]The methodology of how Revelio Labs classify seniority is available at: https://www.data-dictionary.reveliolabs.com/methodology.html#seniority.

To measure AI exposure, I use the LLM exposure score of Eloundou et al. (2024), which estimates the share of task time that could be reduced by at least 50 per cent with access to a large language model. I merge this occupation-level score to Revelio using O*NET–SOC codes, with an 89.5 per cent match rate. The score enters the main regressions continuously. In the extensive-margin mobility sample, it ranges from 0 to 1, with mean 0.494, standard deviation 0.177, and interquartile range 0.147. I therefore interpret mobility coefficients using one-standard-deviation changes as well as full-scale changes. For robustness, I also use the AI Occupational Exposure index of Felten et al. (2021) and the AI Applicability Score of Tomlinson et al. (2025), reported in Appendix B.

The main compositional-shift analysis uses position starts from 2019H1 to 2025H2. I collapse the data into occupation × half-year cells and estimate

$$\text{JuniorShare}_{o,t} = \sum_{k \neq k_0} \beta_k \big(m_{1,o} \times \mathbb{1}[t = k] \big) + \alpha_o + \gamma_t + \varepsilon_{o,t} \tag{6}$$

where $\text{JuniorShare}_{o,t}$ is the share of new hires classified as junior in occupation o and half-year t; $m_{1,o}$ is LLM exposure; α_o and γ_t are occupation and half-year fixed effects; and 2022H2 is the omitted baseline. I use 2022H2 as the reference period because ChatGPT was released late in that half-year, leaving little time for hiring responses to appear in position start dates. I weight cells by the number of positions and cluster standard errors at the O*NET level. I also report a pooled difference-in-differences version with $m_{1,o} \times \text{Post}_t$, where Post_t begins in 2023H1.

For mobility, I construct consecutive position pairs for each worker and define the transition date as the start date of the next position. After cleaning, the dataset contains 12.6 million valid transitions.[3] The mobility post period begins in November 2022.

I separate the mobility analysis into an extensive margin and an intensive margin. The extensive margin asks whether a position is followed by another observed position

[3] I exclude individuals with more than 50 recorded positions (likely data artefacts), drop all positions for individuals with any overlapping employment spells to avoid ambiguous transition timing, and remove positions with missing start dates.

in the panel. The intensive margin conditions on a transition and asks what kind of move occurs. This distinction matters because the model predicts changes in the relative value of junior and senior work, but it does not itself contain a search or matching process.

The extensive margin specification estimates a linear probability model of the form

$$\text{Transition}_i = \delta_1 \text{AI}_o \times \text{Post}_i + \delta_2 \text{Junior}_i \times \text{AI}_o \times \text{Post}_i + \boldsymbol{X}_{i'}\eta + \alpha_o + \gamma_{s(i)} + \varepsilon_i \quad (7)$$

where Transition_i equals one if origin-position observation i is followed by another position; $\boldsymbol{X}_i$ includes lower-order interactions; and $s(i)$ indexes the origin start month. The coefficient δ_1 captures the senior AI-exposure effect after ChatGPT, while δ_2 captures the additional junior effect. Conditional on transition, I estimate seniority gains using

$$\Delta\text{Seniority}_i = \theta_1 \text{AI}_o \times \text{Post}_i + \theta_2 \text{Junior}_i \times \text{AI}_o \times \text{Post}_i + \boldsymbol{X}_{i'}\mu + \alpha_o + \gamma_{\tau(i)} + u_i \quad (8)$$

where $\tau(i)$ is the transition month. I also use log salary change and occupation switching as intensive-margin outcomes.

All mobility specifications include occupation and month fixed effects. The occupation fixed effects absorb permanent differences across occupations, including the level of AI exposure. The month fixed effects absorb common time shocks. Identification therefore comes from whether junior and senior workers in more exposed occupations change differentially after ChatGPT.

Because the intensive-margin regressions condition on transition, I also report a Heckman (1979) two-step correction. The concern is that the set of workers who transition after ChatGPT may not be random. For example, if weaker juniors are displaced and only stronger juniors move successfully, the observed seniority gain could overstate true upgrading. The selection equation uses the main regressors, occupation fixed effects, and a firm-level exclusion restriction: the pre-period mean separation rate at the origin firm, entered directly and interacted with junior status. The logic is that firms with historically high turnover create more transitions, but conditional on moving, their historical separation rate should not directly determine the seniority gain

in the next job. I estimate the probit on a 10 percent user-id subsample and include the inverse Mills ratio in the outcome equation. The corrected standard errors do not include Murphy–Topel or bootstrap adjustment, so they should be read cautiously.

3.2 Identification Strategy

The causal interpretation rests on two conditions. First, exposure must not be shaped by post-treatment labour-market adjustment. This holds by construction for the LLM exposure score, which is based on pre-existing O*NET task descriptions rather than firm adoption decisions. This differs from firm-level AI adoption measures, which may respond to workforce composition.

Second, the timing of the shock should be plausibly unanticipated. Figure 1 shows that public attention to "ChatGPT" was essentially zero before its release on 30 November 2022 and rose sharply afterward. I therefore use 2022H2 as the omitted period in Equation 6 and define the compositional-shift post period from 2023H1 onward.

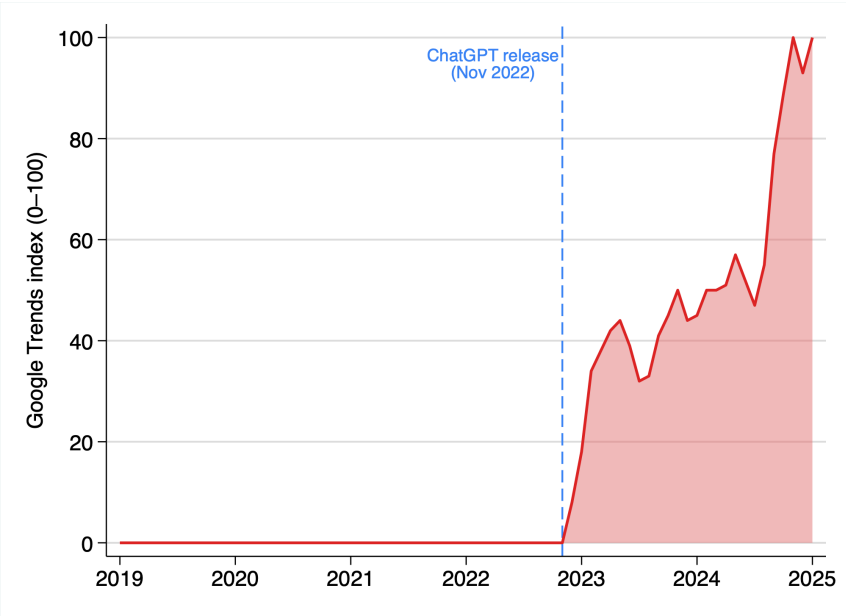

Figure 1: Google Trends index for the keyword "ChatGPT", 2019–2025

The key identifying assumption is parallel evolution of junior hiring shares across exposure levels absent generative AI. I assess this using the pre-treatment event-study coefficients and a formal linear pre-trend test following Freyaldenhoven et al. (2019).

Two limitations remain. I cannot include firm × time fixed effects because treatment varies at the occupation level. Therefore, firm-level shocks correlated with occupation exposure cannot be fully ruled out. I also hold the junior–senior threshold fixed, although AI may change expertise requirements within occupations over time. Finally, the COVID period remains in the pre-period. Half-year and month fixed effects absorb common pandemic shocks, and the event study checks whether exposed occupations were already diverging before ChatGPT.

4 Results

I first document the main compositional result: the junior share of new hires falls in more AI-exposed occupations after ChatGPT. I then examine mobility outcomes to understand how workers adjust after the shock.

This result is related to the firm-level evidence in Klein Teeselink (2025), but the empirical margin is different. That study focuses on firm-level employment outcomes, while my main outcome is the junior hiring share within occupation–half-year cells. The estimates are therefore not directly comparable, but both point to junior workers bearing an important part of the early adjustment.

4.1 Compositional Shift

Figure 2 plots the monthly junior share of new position starts for occupations above and below the median AI exposure score. Before late 2022, both series are stable, with the high-exposure group around 55 per cent and the low-exposure group around 85 per cent. From early 2023, the high-exposure series falls sharply, dropping below 40 per cent by late 2025, while the low-exposure series declines only modestly. The timing is closely aligned with the diffusion of generative AI.

Figure 2: Monthly junior share of new position starts by AI exposure group. Junior defined as seniority ≤ 2 (Revelio Labs classification). AI exposure median split at 0.366 (Eloundou et al. (2024)). Vertical dashed line marks ChatGPT release (November 2022).

The pooled difference-in-differences estimate in Table 2 is -0.096 ($t = -8.71$). Thus, after ChatGPT, occupations with higher exposure experience a larger decline in the junior hiring share. A full-scale increase in exposure corresponds to a 9.6 percentage-point reduction.

	Coefficient	**Std. error**	t	**95% CI**
$\text{AI}_o \times \text{Post}_t$	-0.096	0.011	-8.71	$[-0.117, -0.074]$

Table 2: Pooled difference-in-differences estimate. Dependent variable: junior hiring share at the occupation–half-year level, weighted by cell size. Occupation and half-year fixed effects absorbed. Standard errors clustered at the O*NET code level. Estimation window: 2019H1–2025H2. $N = 12,760$.

The event study in Figure 3 provides the main identification check. The pre-treatment coefficients fluctuate around zero with no clear trend, and the 2022H1 coefficient is small and insignificant (-0.005, $p = 0.44$). There is a mild positive hump during 2019H2–2020H2, plausibly related to COVID-era differences across white-collar occupations, but it disappears before the treatment window. A formal linear pre-trend test following Freyaldenhoven et al. (2019) also gives an essentially zero slope. After 2023H1, the

coefficients turn negative and grow in magnitude: −0.064 in 2023H1, −0.103 in 2024H1, and −0.115 by 2025H2. All post-period estimates are significant at the 1 per cent level. The post-treatment decline is larger and more persistent than the pre-period fluctuations. The quarterly event study in Appendix Figure 6 shows a similar timing pattern.

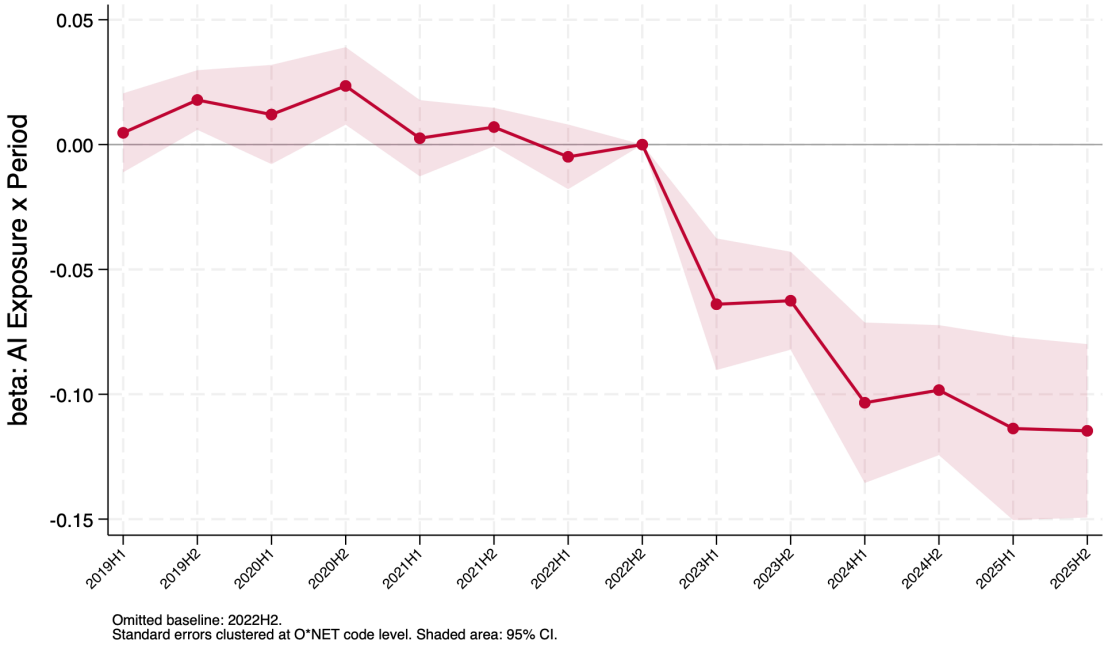

Figure 3: Event study: effect of AI exposure on junior share of new position starts. Each point represents the estimated coefficient $\hat{\beta}_k$ from Equation 6. Estimation window: 2019H1–2025H2. Baseline period: 2022H2. Standard errors clustered at the O*NET code level; shaded area shows 95% confidence interval.

Overall, the evidence supports the first model prediction: high-exposure occupations experience a sustained decline in the junior share of new hires after the diffusion of generative AI.

The magnitude is economically meaningful. By 2025H2, the event-study estimate implies an 11.5 percentage-point lower junior hiring share for a fully exposed occupation relative to the baseline period. Even though the full zero-to-one exposure change is large, the direction and timing are clear: the break occurs after ChatGPT becomes widely known and continues through the post period.

The event-study design is important here because the pooled estimate alone would be hard to interpret if high- and low-exposure occupations were already on different

paths. The absence of a clear pre-trend does not prove the identifying assumption, but it makes the post-period break more credible. It also helps separate the AI-related decline from the temporary COVID-era fluctuations visible earlier in the sample.

4.1.1 Robustness Check

Appendix Appendix B reports robustness checks. The result is similar using the AI Occupational Exposure index of Felten et al. (2021) and under a binary exposure split. The AI Applicability Score of Tomlinson et al. (2025) gives a smaller and insignificant estimate, which is less surprising because it is based on observed usage rather than pre-determined task exposure. This measure may capture adoption frictions as well as exposure. Varying the junior cutoff shows that the effect is strongest at the baseline definition, seniority ≤ 2, and weaker when junior is defined only as level 1 or expanded to level 3. I therefore interpret the baseline result as evidence about workers below the expertise threshold, not about one exact job-title category.

4.2 Career Mobility

I next examine career mobility. These estimates are not direct tests of the static model, but they show how workers adjust after the compositional shift. I study transition probability, seniority gains, log salary changes, and occupation switching.

This worker-level analysis is useful for two reasons. First, a fall in the junior hiring share does not show what happens to workers who are already in exposed occupations. Second, gross flows may reveal reallocation even when net occupation-level employment changes are small. The mobility results should therefore be read as descriptive evidence on adjustment margins rather than as structural estimates of the model.

4.2.1 Extensive Margin: Who Moves?

Table 3 reports the extensive-margin results. The coefficient on AI $\times$ Post is 0.548, implying a 9.7 percentage-point increase in senior transition probability for a one-standard-deviation increase in exposure. This pattern is consistent with higher outside

options and poaching for senior workers, but it could also reflect exits from worse matches. I therefore interpret it as increased gross senior reallocation, not as direct evidence on net senior employment. The triple interaction is -0.660, so the AI-specific mobility gradient is weaker for juniors. Combining the two treatment terms gives about -0.11 per full-scale unit, or -2.0 percentage points for a one-standard-deviation increase. This is a relative AI-exposure effect, not an absolute decline in junior mobility, because the common post-period movement and the junior-specific post effect are also part of the total transition probability. The result is almost unchanged when excluding recent position starts (-0.662), suggesting that short-run right-censoring is not driving the estimate.

This pattern is consistent with the model's junior-displacement logic, but the senior-side interpretation is more cautious. The model predicts higher senior labour demand, whereas the empirical outcome here is a gross transition probability. A senior transition can be a move into a better match, but it can also be an exit from a weaker match. For this reason, I treat the extensive-margin result as evidence of reallocation rather than as a direct test of senior employment growth.

	Full Sample	**Excluding Recent**
Junior	–0.075***	–0.044***
	(0.010)	(0.010)
Junior × AI	0.205***	0.151***
	(0.019)	(0.018)
Junior × Post	0.377***	0.370***
	(0.009)	(0.008)
AI × Post	0.548***	0.567***
	(0.028)	(0.028)
Junior × AI × Post	–0.660***	–0.662***
	(0.031)	(0.029)
N	27,471,965	24,531,422
R^2	0.181	0.158
Occupation FE	Yes	Yes
Month FE	Yes	Yes

Table 3: Career mobility regressions: extensive margin (probability of job transition)

Notes: *Dependent variable is an indicator for whether the worker transitions to a new position. Column (1) uses the full sample; column (2) excludes positions starting after 2024m6 to assess censoring of recently started positions. AI exposure enters as the raw LLM exposure score. In the full-sample estimation sample, its range is 0–1, its standard deviation is 0.177, and its interquartile range is 0.147. All specifications are linear probability models with occupation and month fixed effects. Standard errors clustered at the O*NET occupation level in parentheses.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.2.2 Intensive Margin: Transition Outcomes

Conditional on transition, Table 4 shows larger seniority gains for juniors leaving high-exposure occupations after ChatGPT. The baseline triple interaction is 0.122, meaning that transitioning juniors in more exposed occupations move to higher-seniority roles relative to the comparison groups. The firm separation rate is a strong first-stage predictor of transition, and after Heckman correction the coefficient remains 0.125. The inverse Mills ratio is negative (-0.129), which suggests negative selection on the outcome dimension: workers more likely to transition tend to have smaller seniority gains. The stability of the treatment coefficient is therefore more important than the exact corrected standard error, and it points toward genuine skill-upgrading among transitioning juniors.

The salary and occupation-switching results are weaker. The triple interaction for log salary change is small and insignificant (0.007, $t = 0.51$), even after controlling for seniority gains. The occupation-switching interaction is also small and insignificant (-0.008, $t = -0.61$). Thus, the post-ChatGPT adjustment appears mainly in hiring composition and seniority progression rather than in wage compression or occupational flight.

The occupation-switching null is useful because it weighs against a simple story in which junior workers respond to AI exposure by leaving exposed occupations entirely. Instead, workers who move appear to remain in related occupational space while changing seniority level. This fits the idea that the early adjustment is within occupation, not only between occupations.

The wage null should also be read carefully. The theoretical model predicts wage movements, but Revelio salaries are predicted rather than directly observed compen-

sation. It is therefore possible that true price effects are noisy or delayed in this dataset. The stronger evidence in this paper is on the quantity and composition margins.

	Seniority Gain		**Log Salary Gain**		**Occupation Switch**	
	(1)	(2)	(3)	(4)	(5)	(6)
Junior	1.534***	1.517***	–0.085***	–0.084***	0.059**	0.067**
	(0.049)	(0.049)	(0.014)	(0.014)	(0.027)	(0.027)
Junior × AI	–0.617***	–0.605***	0.031	0.030	–0.034	–0.040
	(0.094)	(0.094)	(0.026)	(0.026)	(0.047)	(0.047)
Junior × Post	–0.114***	–0.117***	0.032***	0.032***	–0.006	–0.004
	(0.015)	(0.015)	(0.007)	(0.007)	(0.007)	(0.007)
AI × Post	0.019	0.035	–0.111***	–0.112***	0.020	0.013
	(0.025)	(0.025)	(0.015)	(0.015)	(0.014)	(0.014)
Junior × AI × Post	0.122***	0.125***	0.007	0.006	–0.008	–0.009
	(0.030)	(0.030)	(0.013)	(0.013)	(0.013)	(0.013)
Δ Seniority			0.203***	0.203***		
			(0.002)	(0.002)		
IMR		–0.129***		0.011***		0.057***
		(0.014)		(0.003)		(0.006)
N	12,487,780	12,487,780	10,676,852	10,676,852	12,487,780	12,487,780
R^2	0.160	0.161	0.350	0.351	0.120	0.121
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Heckman correction	No	Yes	No	Yes	No	Yes

Table 4: Career mobility regressions: intensive margin outcomes conditional on job transition

Notes: *Sample restricted to workers observed in at least two positions. Columns (1)–(2) report seniority gain (levels); columns (3)–(4) report log salary change controlling for seniority gain; columns (5)–(6) report occupation switching (linear probability model). Odd columns are baseline OLS; even columns include Heckman selection correction using firm-level pre-period separation rate as the exclusion restriction. Standard errors clustered at the O*NET occupation level in parentheses. The standard errors in the Heckman columns do not incorporate Murphy–Topel or bootstrap correction for generated-regressor uncertainty and may be understated. Significance stars in those columns should therefore be interpreted cautiously.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5 Discussion

A natural question arising from the results is why I do not detect the wage movements predicted by the theoretical framework. In the model, junior employment falls because w_J falls, and senior employment rises because w_S rises. A high labour supply elasticity ε can make these wage movements small relative to the quantity response, but

it does not remove the price response altogether. The absence of a detected wage effect should therefore be interpreted as a limitation of the empirical test rather than as direct confirmation of the wage predictions of the model. One important caveat is that the salary data from Revelio Labs are predicted values, imputed from job titles, locations, and company characteristics, rather than actual reported compensation. This means that the wage evidence is available, but less reliable than the employment and mobility measures. It is worth noting that Lichtinger & Hosseini Maasoum (2025), who draw on the same Revelio Labs data infrastructure, do not report any wage-based analysis, which may itself reflect an awareness of this limitation. A more precise test of the price channel is therefore an important direction for future research, especially with administrative wage records or other datasets containing reliable individual compensation.

The Heckman correction in the mobility analysis provides a further window into mechanisms beyond the formal model. The estimated inverse Mills ratio is negative ($\hat{\lambda} = -0.129$), indicating that workers with higher unobserved propensity to transition tend to experience smaller seniority gains upon moving. In the standard parameterisation, this implies negative selection on the outcome dimension: the unobservables that make a worker more likely to transition are associated with *worse*, not better, destination matches. The key finding, however, is that the triple-interaction point estimate is virtually unchanged after correction (0.125 versus 0.122 in the baseline). Since the reported standard errors in the corrected columns do not include Murphy–Topel or bootstrap adjustment, the precision of these estimates may be overstated. I therefore place more weight on the stability of the point estimate than on the exact corrected standard error. The preferred empirical interpretation is one of genuine skill-upgrading among transitioning juniors: AI tools may help capable junior workers in exposed occupations build competencies faster, enabling larger seniority gains upon transition. The occupation-switching null reinforces this reading — junior workers in high-exposure occupations are not fleeing to unrelated fields but moving upward within

the same occupational domain. Because the theoretical framework treats junior and senior workers as fixed types, this result should be viewed as a complementary career-progression finding rather than as a formal model prediction.

The extensive-margin results merit further discussion. Senior workers in high-AI-exposure occupations become substantially more mobile after ChatGPT, while the AI-specific mobility gradient is weaker for junior workers. This does not mean that junior transition probabilities fell in absolute terms: the positive $\text{Junior} \times \text{Post}$ coefficient indicates a positive junior-specific post differential relative to seniors, while $\beta_4 + \beta_5$ only captures how this differential changes with AI exposure. One natural concern is whether increased senior mobility should be read as evidence for the model's prediction that senior employment rises ($\partial n_S / \partial A > 0$). The answer is no: the extensive-margin estimate captures gross transitions, not inflows, outflows, or stocks. A senior transition can reflect poaching into a better match, which would be consistent with augmentation. It can also reflect exit from deteriorating conditions in exposed occupations. Distinguishing these mechanisms would require constructing occupation-level inflows, outflows, and net senior headcount changes from the position start and end dates. Such an analysis is feasible with a richer use of the Revelio Labs position data, but lies beyond the scope of the current paper. A related limitation is right-censoring. Excluding very recent position starts shows that the estimates are not driven mechanically by the final months of the panel, but it is a weak diagnostic for censoring that is spread across the post-period. A stronger test would impose a common potential follow-up window, such as retaining only positions with at least 24 months of possible observation, or estimate a duration model that treats ongoing spells as censored. I therefore interpret senior mobility as evidence of increased reallocation among senior workers, not as a standalone confirmation of senior augmentation.

A broader concern for this literature is the lack of consensus on how to measure occupational exposure to AI. Gimbel et al. (2026) document that the most widely used

indices — including those of Eloundou et al. (2024), Felten et al. (2021), and Webb (2020) — disagree substantially on which occupations are most exposed, reflecting differences in whether exposure is defined through task descriptions, technical benchmarks, or patent-based measures. This motivates my use of three methodologically distinct indices: the primary LLM exposure score based on expert and GPT-4 assessment of task-level automation potential, the AI Occupational Exposure index derived from mapping AI capabilities to occupational abilities, and the AI Applicability Score constructed from revealed usage patterns in real-world AI interactions. If the main findings hold across all three, this substantially strengthens the confidence that the results are not driven by a particular measurement choice. A separate data limitation concerns the representativeness of the Revelio Labs sample. Because the data are drawn from public LinkedIn profiles, the sample likely over-represents white-collar, professional occupations relative to the working population as a whole. Since LLM exposure is concentrated in precisely such occupations, this sample composition may amplify the estimated effects relative to what a nationally representative survey would yield.

The AI Applicability Score also deserves a cautious interpretation. Unlike the Eloundou and Felten measures, it is based on observed Copilot use during the post-treatment period. This makes it useful as a revealed-use comparison, but weaker as an exogenous exposure measure. Its weaker result therefore does not overturn the main finding based on pre-determined task exposure.

Finally, as noted in Section 3, the seniority threshold separating junior from senior workers is held fixed throughout the analysis, yet AI may itself reshape expertise requirements within occupations, shifting this boundary endogenously over time. Modelling such a response would require the between-occupation general equilibrium framework of Autor & Thompson (2025) and is left for future work. More broadly, the estimates presented here capture the early stages of generative AI diffusion. The capabilities of the technology are expanding rapidly — from text-based assistance toward autonomous

task execution — and the occupational task bundles that define current exposure scores may not remain stable as these capabilities evolve. Whether the compositional shifts documented in this essay persist, intensify, or reverse as firms and workers adapt over a longer horizon remains an open question, and one that longer post-treatment panels will be well placed to address.

6 Conclusion

This essay has argued that aggregate null results for AI's employment effects conceal meaningful compositional shifts within occupations. Extending the expertise framework of Autor & Thompson (2025) to permit within-occupation task specialisation, I derive formal predictions for how AI capability improvements affect junior employment shares, senior marginal products, wages, and net occupation-level employment. The evidence is strongest for the model's junior compositional prediction: in occupations more exposed to large language models, the junior hiring share declines. The worker-level mobility results provide complementary evidence on adjustment margins beyond the static model. Juniors who transition achieve larger seniority gains, consistent with genuine skill-upgrading, while senior workers become substantially more mobile. These results should be interpreted as career-progression and gross-reallocation patterns rather than direct tests of formal model predictions. By contrast, I do not detect the wage movements predicted by the formal model in the Revelio predicted salary data, so the price-margin results should be interpreted cautiously.

These findings carry a subtler implication. Much of the public debate frames AI as a threat to entire occupations or skill groups. The evidence here suggests a different pattern: AI reshapes *who does the work within an occupation* before it reshapes *whether the occupation exists*. If this pattern generalises, then the workers most immediately affected are not those in "threatened occupations" but those at the bottom of the ladder

within occupations that are, on the whole, thriving. This is a form of displacement that aggregate statistics are poorly equipped to detect, and that conventional policy responses — retraining workers for different occupations — are poorly designed to address. The more relevant question may be how firms and institutions restructure entry-level pathways when the tasks that once served as training grounds are increasingly performed by machines.

Whether these early compositional shifts prove transitory or permanent, and whether the expertise threshold itself adapts in response, are questions that longer post-treatment data will help resolve. But if displacement does begin at the junior level, as the evidence here suggests, the implications extend beyond the immediately affected workers. Every senior professional was once a junior one; entry-level positions have traditionally served as the channel through which new labour market entrants acquire occupation-specific skills and progress toward independent judgment. A sustained contraction of this channel would warrant attention from policymakers and educational institutions, not because entire occupations are disappearing, but because the pathway into them is narrowing.